Reaction Times Analysis Pipeline

Problem Description

Understanding how users draw objects and how their reaction time changes across drawing attempts can provide insights into user behavior and cognitive processing. Drawing datasets often contain thousands of records describing user interactions, including the object being drawn and the time taken to respond.

However, these datasets are large and difficult to analyze directly in their raw form. Efficient analysis requires extracting the data, transforming it into aggregated forms, and visualizing patterns in user behavior.

This project builds a small data pipeline that extracts drawing data from a cloud data warehouse, processes it using analytical queries, and produces visualizations for analysis. The pipeline enables two key analyses:

- **Concept distribution:** identifying which objects are drawn most frequently.
- **Reaction time trends:** observing how average reaction time changes across drawing trials.

These analyses provide insights into both the popularity of different drawing concepts and how user response times evolve as they complete multiple drawing attempts.

Cloud Infrastructure

The project uses **Google Cloud Platform (GCP)** as the cloud environment for data storage and extraction.

The drawing dataset is stored in **BigQuery**, Google's fully managed cloud data warehouse. BigQuery provides scalable storage and SQL-based querying capabilities for large datasets.

To retrieve the data for analysis, the **BigQuery command-line interface (`bq`)** was used within **Google Cloud Shell**. The dataset was exported from BigQuery as a CSV file and downloaded for downstream processing.

Using Cloud Shell allows the analysis workflow to run directly within the cloud environment without requiring additional local setup. The `bq` CLI acts as a reproducible interface for exporting data from BigQuery tables.

Data Ingestion (Batch Extraction)

The pipeline uses a **batch ingestion approach** to extract data from the warehouse.

The ingestion process consists of the following steps:

1. Query the drawing dataset stored in BigQuery.
2. Export the query results using the `bq` command-line tool.
3. Save the results as a CSV file (`drawings.csv`).
4. Use the exported CSV as the input dataset for downstream analysis.

Example workflow:

BigQuery Dataset

↓

bq query / bq extract

↓

drawings.csv

↓

Local analysis pipeline

This approach allows the dataset to be extracted efficiently from the cloud warehouse and reused for repeated analysis without repeatedly querying BigQuery.

Data Warehouse

For analytical processing, the project uses **DuckDB**, an embedded analytical database optimized for OLAP workloads.

DuckDB allows SQL queries to run directly on CSV files without requiring a traditional database server. The exported dataset (`drawings.csv`) is loaded into DuckDB and queried to compute aggregated metrics.

Two analytical tables are generated:

concept_distribution

Counts the number of drawings associated with each concept.

Example query:

```
SELECT
    concept,
    COUNT(*) AS drawing_count
FROM drawings
GROUP BY concept
ORDER BY drawing_count DESC
```

reaction_time_over_trials

Calculates the average reaction time for each trial index.

Example query:

```
SELECT
    trial_index,
    AVG(rt) AS avg_reaction_time
FROM drawings
GROUP BY trial_index
ORDER BY trial_index
```

These tables organize the dataset into aggregated structures that support efficient analysis and visualization.

Transformations

Data transformations are implemented using **SQL queries executed through DuckDB in Python**.

The transformation pipeline performs:

1. Loading the CSV dataset into DuckDB
2. Aggregating drawing counts by concept
3. Computing average reaction times by trial index
4. Exporting the results to CSV files

The transformation logic is implemented in the script:

generate_analysis.py

This script executes the SQL queries and generates the aggregated datasets used for visualization.

Dashboard

The results are visualized using **Power BI**.

Two visualizations are generated from the processed datasets:

Concept Distribution

A bar chart showing the most frequently drawn objects.

Reaction Time Over Trials

A line chart showing how average reaction time changes as users complete additional drawing trials.

These visualizations provide an intuitive view of the dataset and highlight patterns in user behavior.

Reproducibility

The project is designed to be reproducible so that others can run the analysis pipeline and generate the same results.

Steps to reproduce the project:

1. Clone the repository

```
git clone https://github.com/Mhanna112-code/Reaction-Times-Analysis.git

cd Reaction-Times-Analysis
```

2. Install required Python libraries

```
pip install duckdb pandas matplotlib
```

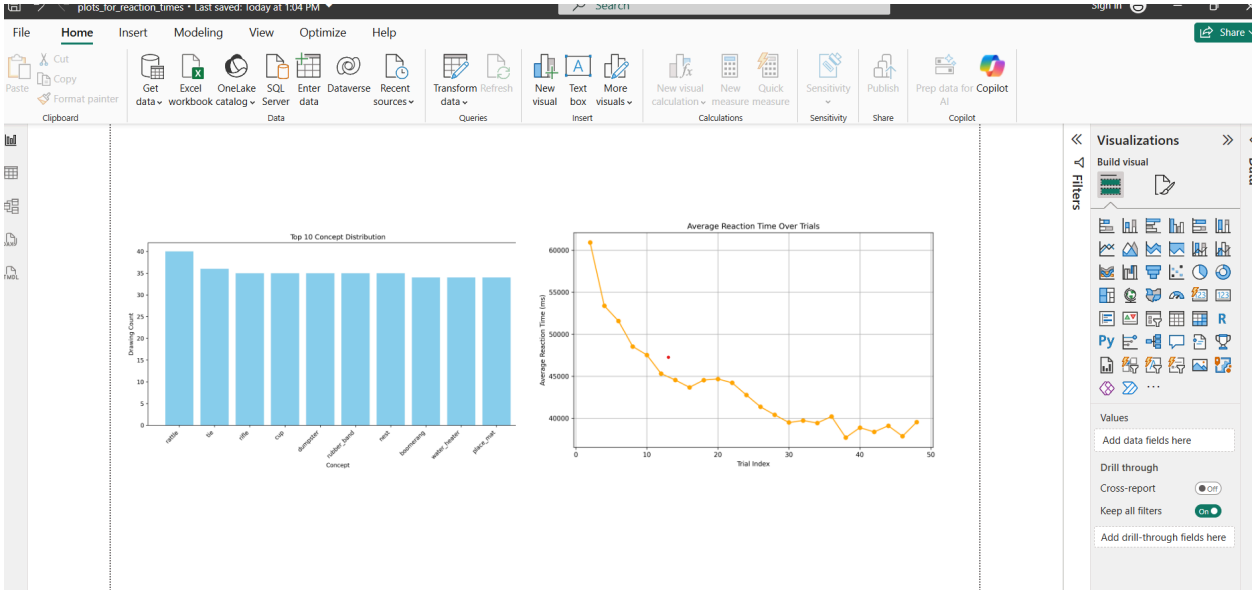
3. Ensure the dataset (`drawings.csv`) is present in the project directory.
4. Run the analysis script

```
python generate_analysis.py
```

5. The script will generate:

- `concept_distribution.csv`
- `reaction_time_over_trials.csv`
- `top_10_concept_distribution.png`
- `reaction_time_over_trials.png`

These .png outputs were imported into Power BI for dashboard visualization:



Technologies Used

- **Google BigQuery** – cloud data warehouse for storing the dataset
- **bq CLI** – extraction of dataset from BigQuery
- **DuckDB** – analytical database for processing the dataset
- **Python** – transformation pipeline
- **Matplotlib** – visualization generation
- **Power BI** – interactive dashboard